OpenFOAM CPU research: hypotheses, measurements and decisions

Orange = NSCC CPU · Blue = Iris CPU · lower seconds/step is better

21 Oct · Iris
Learn the actual cluster route
Interactive SSH jump → native MPI chain
Jobs 2293751–2293754
Full mesh: 21.9654 s/step; 40 steps
PRACTICE / not official score

21 Oct · NSCC
Establish an unchanged reference
Full preparation before solver timing
Jobs 25905114 / 25907325
Baseline 14.9220 s/step; ordered Scotch pair −0.994% (noisy)
REFERENCE REPRODUCTION

motivated

21 Oct · NSCC
Test NUMA-aware decomposition
Change Scotch grouping coefficients
Jobs 25907688 / 25907753
Graph cut −10.63%; solver time +3.44%
CLOSED / no stable gain

21 Oct

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21–22 Oct · NSCC
Separate a graph proxy from runtime
Compare N256 partition interfaces
Jobs 25907772
Preparation only; no solver speed ranking
MECHANISM / not a speedup

22 Oct

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22–23 Oct · NSCC
Sweep rank density and node count
Six four-node native logs reviewed
Jobs 25921488 and sibling logs
Scotch-128: 3.6779 s/step; 150 steps
OBSERVATION / separate allocations

repeat

23 Oct onward · NSCC
Reproduce the organizer recipe
4 nodes · 512 ranks · full 150 steps
Jobs 25940700
3.364622070 s/step; baseline reproduction
REFERENCE REPRODUCTION

23 Oct

27–28 Oct · Iris
Test hierarchical axis order
zyx versus xyz; full-length reverse order
Jobs 2294234 / 2294262
zyx 21.2296 then 26.6108 s/step; first arm wins
CLOSED / no stable gain

27–28 Oct

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29 Oct · Iris
Move communicating ranks closer
Graph-based rank-to-core placement; reverse cycle
Jobs 2294390
Best arm 21.0512 (14 steps); pooled +5.37% slower
CLOSED / no stable gain

29 Oct

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1 Nov · Iris
Isolate a supplied runtime setting
Collated versus uncollated I/O; reverse order
Jobs 2294635 and parent pair
Apparent −6.07% reverses to +9.35%
CLOSED / no stable gain

1 Nov

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1–2 Nov · Iris
Port teammate settings fairly
Same Scotch source; socket/core MPI + collated I/O
Jobs 2294637
24.7453 vs 27.8735 s/step; −11.22%
QUALIFIED SIGNAL / repeat needed

control question

2 Nov · Iris
Compare our partition with Scotch
Plain zyx versus its matched organizer control
Jobs 2294640
29.6490 vs 26.9011 s/step; +10.21%
CLOSED / no stable gain

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2 Nov · Iris
Check a new axis order and reverse it
40-step screen → 52-step confirmation
Jobs 2294643 / 2294644
−0.71% initial signal → +10.25% slower
CLOSED / no stable gain

2 Nov

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3 Nov · Iris
Revisit native xyz with a control
20-step four-arm screen
Jobs 2294671
Second candidate timed out; comparison incomplete
INCOMPLETE / invalid comparison

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3 Nov · Iris
Test preparation memory policy
Inherited / interleaved child; copy + hash only
Jobs 2294674 failed → 2294675 qualified
File-page counter split changes; no solver result
MECHANISM / not a speedup

3 Nov

mechanism → solver test

5 Nov review (Vietnam) · Iris
Check whether the preparation-policy gain repeats
Same Scotch; only preparation-child policy changes
Jobs 2294676 /2294735 /2294800
Independent ABBA 40: 29.0048 versus 27.3379 s/step; +6.10% slower
CLOSED / no stable gain

5 Nov

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5 Nov · Iris
Test solver-rank interleave policy
Qualify outer and in-rank numactl behavior, then compare default versus in-rank policy in a 40-step ABBA solver trial.
Jobs 2294801 / 2294802 / 2294803
Control A1 completed at 24.7586 s/step; candidate B1 timed out after 33 timings (partial mean 41.8465 s/step; not a score). Remaining arms did not start.
INCOMPLETE / invalid comparison

motivated

5 Nov review (Vietnam) · Iris
Qualify tiny Allreduce before a solver trial
Default MPI versus forced tuned recursive doubling; 8/64-byte correctness-checked ABBA probes
Jobs 2294805
8B block 1: default p50 2.478 us; forced tuned 4.117 us; candidate fails benefit gate
CLOSED / no stable gain

5 Nov